NOTE

Paleomagnetism of dykes from the Groundhog River Block, northern Ontario: implications for the uplift history of the Kapuskasing Structural Zone[1]

M. P. BATES AND H. C. HALLS
Department of Geology, University of Toronto, Erindale Campus, Mississauga, Ont., Canada L5L 1C6

Received November 20, 1990
Revision accepted April 5, 1991

The groundhog River Block (GB) forms the central segment of the Kapuskasing Structural Zone a fault-bounded belt of Proterozoic crustal uplift in the southern Superior Province. Dykes adjacent to the GB carry a magnetization ($D = 11°$, $I = 26°$, $\alpha_{95} = 27°$, $N = 3$) that is characteristic of 2.45 Ga Matachewan dykes. However, within the GB, dykes and high-grade gneissic host rocks carry a steep positive magnetization ($D = 295°$, $I = 83°$, $\alpha_{95} = 12°$, $N = 9$) that also partially overprints the dykes immediately outside the GB . The boundaries between the contrasting paleomagnetic signatures coincide with boundary faults and changes in paleopressure that define the GB. The steep magnetization in the GB was acquired during uplift and cooling close to the time at about 1.95 Ga when a virtually antiparallel remanence of similar origin was being formed in the Chapleau Block, the southern segment of the Kapuskasing Structural Zone.

Le bloc de Groundhog River (BG) représente le segment central de la Zone structurale de Kapuskasing, une ceinture bordée par des failles d'un soulèvement crustal protérozoïque dans le sud de la province du lac Supérieur. Les dykes adjacents au BG portent une aimantation ($D = 11°$, $I = 26°$, $\alpha_{95} = 27°$, $N = 3$) caractéristique des dykes de Matachewan âgés de 2,45 Ga. Cependant, dans le BG, les dykes et les gneiss de degré élevé de métamorphisme encaissants portent une aimantation positive fortement inclinée ($D = 295°$, $I = 83°$, $\alpha_{95} = 12°$, $N = 9$) qui aussi surimprime partiellement les dykes présents dans l'entourage immédiat du BG. Les limites qui séparent ces deux signatures paléomagnétiques contrastantes coïncident avec les failles et les changements de paléopression définis par le BG. L'aimantation fortement inclinée dans le BG fut acquise durant les évènements de soulèvement et de fermeture de refroidissement, il y a environ 1,95 Ga, temps correspondant à l'acquisition d'une rémanence virtuelle anti-parallèle d'origine similaire dans le bloc de Chapleau, le segment méridional de la Zone structurale de Kapuskasing.

[Traduit par la rédaction]

Can. J. Earth Sci. **28**, 1424–1428 (1991)

Introduction

The Kapuskasing Structural Zone (KSZ) is a discontinuous band of upper-amphibolite to granulite-grade gneiss (Percival and Card 1985) traversing the lower-grade Superior Province granite–greenstone terrain from Lake Superior to James Bay (Fig. 1). KSZ uplift occurred at approximately 1.95 Ga (Percival *et al.* 1988) along west-dipping thrusts (Cook 1985) of the Ivanhoe Lake Cataclastic Zone (ILCZ), exposing a cross section of the middle crust to a depth of more than 20 km (Percival and Card 1985; Percival 1988).

The KSZ consists of three fault-bounded blocks: Fraserdale–Moosonee, Groundhog River, and Chapleau (FB, GB, CB), from north to south, respectively (Percival and McGrath 1986). Most current and past research has focused on the CB, the largest, best-exposed, and most accessible of the three blocks. Conversely, the GB has attracted less attention, and the FB is poorly known.

In the CB, two dyke swarms, the 2.45 Ga Matachewan and 2.0 Ga Kapuskasing, have been identified (Halls and Palmer 1990), but only the latter dykes are found in the highest granulite grade rocks of the block. A notable feature of dykes in both swarms is that compared with dykes outside the CB, they are relatively fresh, contain amphiboles with relatively high alumina (Palmer and Barnett 1988), and carry feldspars that show a distinct tea-coloured clouding (Halls and Palmer 1990). These features are thought to reflect crystallization at relatively deep crustal levels (Halls and Palmer 1990). A further feature of the Matachewan dykes is that they have a "normal" polarity of magnetization in contrast with dykes of the same swarm outside the KSZ which have a dominantly "reversed" magnetization (Halls and Palmer 1990; Bates and Halls 1990). This change in polarity is a consequence of uplift within the KSZ, exposing deeper levels of dykes that acquired their magnetization in a younger "normal" polarity epoch (Halls and Palmer 1990; Bates and Halls 1990).

The absence of Matachewan dykes within Chapleau Block granulites contrasts with the occurrence of dykes with a similar trend to the Matachewan swarm in the granulites of the GB (LeClair and Nagerl 1988). In view of the potential vertical zonation of magnetic polarity, petrology, and geochemistry of the Matachewan dykes, we have carried out a paleomagnetic and petrological examination of the GB dykes to see if they resemble those in the Chapleau Block and have acquired an uplift-related remanence similar to that found in the Kapuskasing swarm (Halls and Palmer 1990).

Sample localities

Oriented samples were collected at 17 sites (Table 1; Fig. 1) from dykes, gneissic country rock, and pseudotachylyte within and bordering the GB. Exposure is limited due to widespread

[1]Lithoprobe Phase II Publication 220; Publication 34 of International Geological Correlation Program (IGCP) Project 257 on Precambrian dyke swarms.

TABLE 1. Description of paleomagnetic sample sites

Site	Strike/dip of dyke/foliation (°)	Dyke width (m)	Paleopressure (kbar)
1	346/90	>11	8.6
2	310/45SE	—	5.9–7.2
3	355/—	9	5.9–7.2
4	028/—	32	5.9–7.2
5	066/—	>2	5.9–7.2
6	354/80E	22	5.9–7.2
7E	354/—	27	5.0
7W	013/90	2	5.0
8	000/50W	5	6.7
9	000/—	25	6.7
10	342/56W	10	6.7
11	344/90	>30	5.5
12	002/90	>22	5.5
13	000/—	17	5.5
14	183/80W	—	5.9–7.2
15	255/90	—	5.9–7.2
16	001/—	>30	—
17	014/—	>17	—

NOTES: All sites are in dykes except sites 14 and 15 (granulite gneiss) and 2 (pseudotachylyte). Site 7 comprises two dykes: the western one is altered and contains pseudotachylyte veins; the eastern one is fresh. Paleopressure estimates (1 kbar = 100 MPa) are after LeClair (1990).

glacial deposits, and the majority of samples were obtained by field drilling of low, flat exposures. Road cuts at sites 9 and 10 and a lakeshore exposure at site 12 allowed block sampling to be employed.

Irregular networks of anastomosing pseudotachylyte veins occur in about half the sites cutting and therefore postdating both gneisses and dykes (sites 2, 4, 6, 7W, 8, 10, and 13). The pseudotachylyte probably formed in association with the main phase of KSZ uplift (Percival 1983; LeClair and Nagerl 1988). Therefore intrusion of the dykes is likely to predate the formation of the GB.

Dyke petrography

Almost all sampled dykes are quartz diabase, with major plagioclase and pyroxene and subsidiary biotite, amphibole, and opaques; only one dyke, at site 3, contains trace olivine. The dominant variety is generally fresh, has blue-green amphibole occurring as discrete grains and as alteration rims around clinopyroxene, and contains feldspars with tea-coloured clouding. The dykes showing these characteristics (sites 1, 3, 6, 7E, and 9–13) all have NW–N trends (mean 350 ± 7°, $N = 9$) and occur both within and adjacent to the GB. The remaining dykes in the same region (sites 4, 5, 7W, and 8) have a more variable, N–NE trend, are more pervasively altered, and have clear feldspars. In the extreme east, remote from the GB, dykes at sites 16 and 17 are strongly altered with clear feldspars. Pseudotachylyte veins cutting the dykes are conspicuous at sites 5 and 8, and microfaults, possibly occupied by pseudotachylyte, are found in the dykes at sites 10–13. Although the dykes cannot be assigned to a swarm on the basis of petrography alone, their N–NW and N–NE trends (Table 1) suggest affiliation, respectively, with the Matachewan swarm and with 2.1 Ga Preissac dykes (Goodwin 1972; Hanes and York 1979) that form the eastern part of an extensive quartz diabase swarm that occurs on both sides of the KSZ.

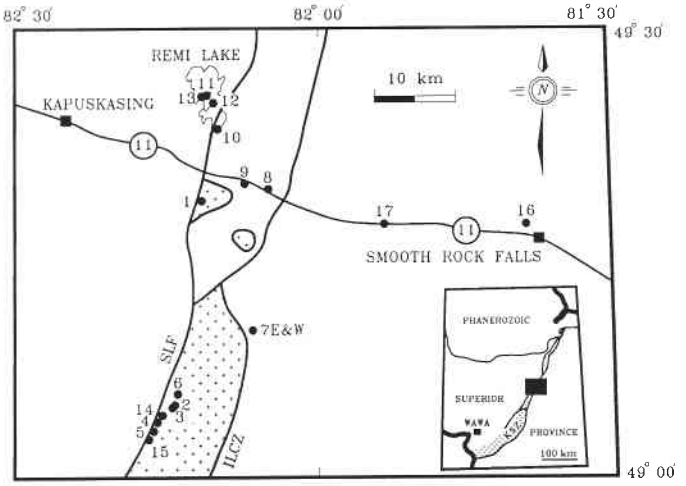


FIG. 1. Sample sites in and around the northern Groundhog River Block. The location of the study area with respect to the Kapuskasing Structural Zone (KSZ) is indicated by the inset. Fault traces (heavy lines) and extent of granulites (cross pattern) are after Percival and McGrath (1968) and LeClair and Nagerl (1988). SLF, Saganash Lake Fault; ILCZ, Ivanhoe Lake Cataclastic Zone.

Anisotropy of magnetic susceptibility (AMS)

The AMS ellipse was obtained for each sample using a Sapphire SI-1 instrument. The degree of anisotropy of low field susceptibility of all rock types is low (k_{max}/k_{min} usually < 1.1), and the principal axes of ellipses are scattered and not related to the local fabric in the gneiss or the shape of the dyke. Therefore the magnetic field does not appear to have been deflected by a strong anisotropy during remanence acquisition. Values of apparent susceptibility (Abrahamsen 1986) are also low (maximum 0.2 SI units), so any deflection into the plane of the dyke is less than 5°.

Paleomagnetism

Samples were subjected to stepwise alternating field (AF) and thermal demagnetization using Schonstedt TD-1 and TSD-1 instruments and measurement on a modified Digico Spinner Magnetometer. Components of remanent magnetization were separated using principal component analysis (Kirschvink 1980), remagnetization circles (Bailey and Halls 1984), and orthogonal vector plots (Zijderveld 1967).

Typically, two linear segments of similar direction are resolved during both AF and thermal demagnetization (Figs. 2*a*–2*d*). The first component removed has a low coercivity range (≤25 mT) and low to intermediate unblocking temperatures (50–530°C). Site means are scattered around the present Earth's field direction (PEF: 349°/73°; Garland 1979), which lies within the circle of 95% confidence of the overall mean (12°/80°, α_{95} = 11.4°). Therefore, we interprete this component as a recent viscous remanent magnetization (VRM).

After removal of the VRM, a steep positive magnetization **A** (Fig. 3*a*) is isolated at all sites and in all lithologies, with a coercivity range of 10–90 mT and a thermal range of 150–580°C. Although similar in direction, the VRM and **A** components are consistently resolved in each sample demagnetized, and **A** has a mean direction (Table 2) with an α_{95} that does not enclose the PEF. For these reasons, we suspect that **A** is not a VRM, but an older, geologically significant remanence.

In some examples of two-component remanence, a third component is implied by the change in direction of the magnetization away from the **A** component toward shallower inclina-

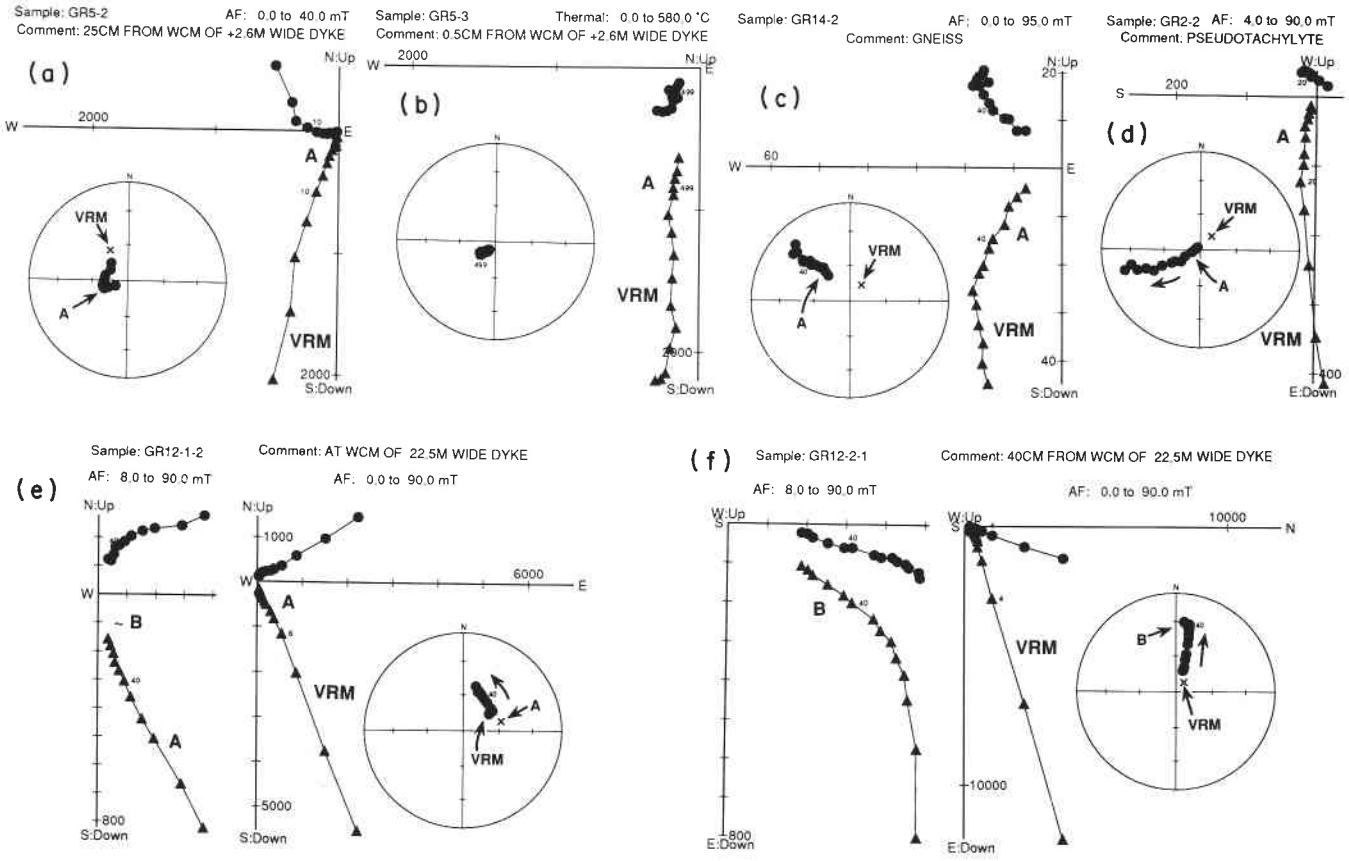


FIG. 2. Examples of demagnetization data displayed on orthogonal vector plots (●, horizontal projection; ▲, vertical projection; units are mA/m) and equal-area stereographic projections (all directions lower hemisphere; direction of interpreted components indicated). (*a*, *b*) Two-component remanence (VRM and **A**) resolved by AF and thermal demagnetization of a diabase dyke. (*c*) Two-component remanence (VRM and **A**); AF demagnetization of high-grade gneiss. (*d*) Two-component remanence (VRM and **A**); AF demagnetization of pseudotachylyte. (*e*) Three-component remanence (VRM, **A**, and **B**); AF demagnetization of a diabase dyke. (*f*) Well-defined linear segment for **B** from the same site.

tions (e.g., Figs. 2*c*, 2*d*). However, no end point is established, and the diverging demagnetization paths either suggest the component is scattered or approximately antipodal to **A**.

In some dykes a high coercivity (20–90 mT), high unblocking temperature (450–580°C) component **B** (Fig. 3*b*, Table 2) can be isolated after the removal of **A** (Fig. 2*e*) or the VRM (Fig. 2*f*). It is defined on the basis of linear segments on vector diagrams (e.g., Fig. 2*f*) and remagnetization circles. Although **A** has intermediate coercivity and unblocking temperature between VRM and **B**, it is well defined by linear segments. Furthermore, where all three components are present (e.g., Fig. 2*e*), demagnetization data do not lie in a single plane and therefore **A** is not a hybrid of the VRM and **B** remanences.

Origin of the A and B magnetizations

Dykes within the GB carry only the steep positively inclined remanence **A** which is also found in host-rock gneisses remote from dykes and in pseudotachylyte which is observed to cut the dykes. This remanence is thus interpreted to be a remanence acquired during uplift within the GB. Geobarometry (LeClair 1990) gives higher paleopressure estimates within the GB (approx. 6.5 kb) than outside (approx. 5 kb, Table 2), suggesting that uplift within the GB was significantly higher than outside and probably sufficient, given a paleodepth of about 25 km and a nonorogenic geothermal gradient of about 25°C/km (e.g., Philpotts 1990; p. 5), to completely reset any remanence carried by magnetite (Curie temperature 580°C) upon subsequent uplift. Even in those parts of the GB where the metamorphic grade of the host rocks is less than granulite, paleopressures are still higher than outside the block which may account for sites 8–10 all showing the uplift-related magnetic overprint **A** to the exclusion of **B**.

Magnetization **B** is similar to the normal polarity remanence found in the Matachewan swarm (e.g., Halls and Palmer 1990) which was formed at the time of dyke emplacement (Irving and Naldrett 1977; Buchan *et al.* 1990). All four dykes carrying this component lie immediately outside the GB and belong to the petrologically fresh group with blue-green amphibole and clouded feldspars. Elsewhere, populations of dominantly normal polarity Matachewan dykes, either within the Chapleau Block or west of the KSZ along Highway 101 (Halls and Palmer 1990) also have clouded feldspars, having been brought up from deeper crustal levels. We therefore suggest that the Matachewan dykes adjacent to the GB have been uplifted during the formation of the KSZ but not enough to totally reset their primary magnetic signature because each dyke shows coexisting **A** and **B** remanences. The petrographic similarity of most north-northwest-trending dykes within the GB suggests that they are also Matachewan dykes but lack a primary paleomagnetic signature.

Acquisition of the **A** magnetization is therefore pervasive in the areas of greatest uplift in the GB, and only partial on its immediate flanks. Furthermore, in an extensive paleomagnetic study of the Matachewan swarm involving more than 300 dykes, no similar steep, stable remanence has been found either to the

TABLE 2. Paleomagnetic results

Site	N	N_C	N_R	D (°)	I (°)	k	α_{95} (°)
				A magnetization			
1[a]	6	—	—	352.7	38.2	43	10.3
2[b]	5	—	2	241.9	83.3	310	4.4
3[b]	3	4	—	155.6	65.8	19	20.0
4[b]	5	—	1	232.4	81.9	11	24.3
5[b]	5	1	—	236.5	73.4	280	4.6
6[b]	5	1	—	317.5	66.5	21	17.0
7E[b]	4	1	—	335.1	59.6	19	21.0
7W[c]	3	—	1	343.2	82.4	7	48.8
8	2	—	—	274.7	73.5	66	31.2
9	3	—	3	349.1	74.4	130	10.9
10[b]	5	—	—	224.3	72.4	26	15.3
11[b]	6	—	—	1.3	70.7	38	11.0
12	3	—	2	19.2	70.0	12	38.0
13[b]	5	—	1	353.6	74.6	65	9.6
14	4	—	1	336.9	71.0	34	15.9
15	3	—	2	48.3	74.7	73	14.5
Mean	9	—	—	294.7	82.6	12	12.3
				B magnetization			
7E	2	3	—	28.3	23.5	12	36.0
11[b]	2	3	1	18.7	35.9	160	11.0
12[b]	4	1	1	15.2	33.8	38	14.0
13[b]	5	1	—	360.0	8.7	46	11.0
Mean	3	—	—	10.6	26.4	21	27.4

NOTES: N, number of samples that yielded stable end points or linear segments on vector plots, or number of sites; N_C, number of samples that yielded remagnetization circles; N_R, number of rejected samples; D, declination; I, inclination; k, precision parameter; α_{95}, semiangle of the cone of 95% confidence.

[a]The direction at this site could represent either an **A** or a rotated **B** remanence.

[b]Sites with $N + N_C \geq 5$ and $\alpha_{95} < 25°$ are included in the final means. Sites 16 and 17 were paleomagnetically unstable.

[c]At this site a second magnetization component has a direction (243°/65.4°, $N = 2$, $N_C = 2$; $k = 62$, $\alpha_{95} = 17°$) that is similar to that found in Preissac dykes (Buchan and Halls 1990), supporting the correlation based on dyke trend.

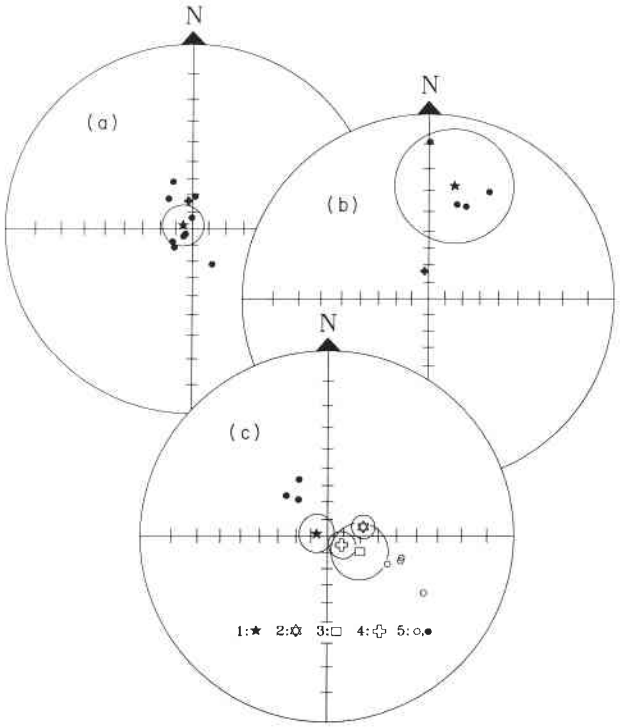


FIG. 3. Equal-angle projections summarizing the paleomagnetic data; closed and open symbols are downward- and upward-pointing magnetizations, respectively. (*a*, *b*) Site directions used to calculate, respectively, the final **A** and **B** means (stars with α_{95} circles). Crosses indicate present Earth's field direction. (*c*) Comparison of the mean **A** direction with existing data from the CB, with α_{95} circles. Symbols: 1, mean **A**, geographically corrected to the CB; 2, KSZ gneisses and Shawmere anorthosite (Symons and Vandall 1990); 3, Shawmere anorthosite (Costanzo-Alvarez and Dunlop 1988); 4, unrotated Kapuskasing dykes (Halls and Palmer 1990); 5, preliminary data for carbonatites of the southern KSZ (Dunlop *et al.* 1988; α_{95} not available).

east or the west of the KSZ (H. C. Halls and M. P. Bates, unpublished data). The spatial relationship recorded here suggests a genetic link between the origin of **A** and the development of the KSZ.

The remagnetization pattern observed in the GB appears to be repeated in the CB where a remanence opposite in polarity to **A** was acquired upon uplift and cooling of the KSZ at about 1.95 Ga (Fig. 3*c*). The CB uplift remanence occurs throughout the granulite-grade rocks of the CB (Symons and Vandall 1990) and has remagnetized the quartz-diabase Kapuskasing dykes, so they yield negative contact tests (Vandall 1986; Halls and Palmer 1990).

It is unlikely that a well-defined uplift reamence is preserved in the CB but totally absent just to the north in granulites and pre-uplift dykes of the GB. Therefore, we suggest that **A** represents the equivalent KSZ uplift remanence in the GB, but acquired at a time of opposite polarity in the ambient magnetic field. The presence of **A** in pseudotachylyte (believed to form at depths of about 10 km; Sibson 1975) at sites 2 and 3 suggests that some of it was acquired at relatively shallow depths, perhaps in part by chemical means.

An alternative theory is that **A** is related to the 1.0–1.1 Ga carbonatite complexes that occur along the trace of the KSZ. Steeply inclined magnetizations, similar to **A** but of both positive and negative inclination, occur in the carbonatites of the CB (Fig. 3*c*) which have chemically remagnetized the surrounding country rocks during emplacement after the main phase of KSZ uplift (Costanzo-Alvarez and Dunlop 1988; Dunlop *et al.* 1988). The nearest carbonatites to the GB are the 1075 Ma Clay Howells complex (Lewchuk and Symons 1988) and the 1890 Ma Cargill carbonatite (U/Pb; L. M. Heaman, personal communication, 1990), but they are at least 80 and 55 km, respectively, from our sampling sites. The Clay Howells complex has a remanence (294.2°/27.1°; Lewchuk and Symons 1988) that is distinctly different from **A**. No paleomagnetic remanence has yet been described from the older Cargill carbonatite west of the GB, but remagnetization due to this intrusion would need to be widespread and does not explain the prominence of the **A** magnetization within the GB. Rather, we would expect the dykes west of the GB to be remagnetized to the same degree as those within it, which is not the case.

Therefore, **A** is interpreted as a remanence acquired upon uplift, although the time of GB uplift was not exactly contemporaneous with that of the CB because the magnetizations are of opposite polarity. The timing of uplift along the KSZ awaits detailed isotopic studies in the GB to compliment those carried out in the CB (Hanes *et al.* 1988; Percival *et al.* 1988).

Conclusions

The granulite-grade gneisses and accompanying dykes of the Kapuskasing Structural Zone acquired a steep paleomagnetic remanence upon uplift of the Groundhog River Block, close to but not exactly coincident with uplift of the Chapleau Block at 1.95 Ga. The difference in timing of uplift between the blocks is reflected in the opposite polarity of the remanence, but as yet we are unable to determine the relative order of uplift. Four Matachewan dykes in the adjacent lower grade rocks of the Superior Province retain a normal polarity primary remanence but are heavily overprinted by the uplift remanence close to the Groundhog River Block boundary faults.

Acknowledgments

A. D. LeClair and J. A. Percival (Geological Survey of Canada) helped in identifying potential sample sites; J. Wong, S. Pfleiderer, and C. Dodds (all University of Toronto) acted as field and laboratory assistants; and G. W. Pearce (Queen's University, Kingston) gave advice on the analysis of anisotropy of magnetic susceptibility. This research was financed by grants from the Lithoprobe Supporting Geoscience Projects Programme and by Natural Sciences and Engineering Research Council of Canada operating grant A7824.

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